Hw9 con.

Inspection: C51 Method 2:

$$S = \left\{ \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 5 \\ -1 \\ 3 \end{bmatrix} \right\}$$

$a = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$

1. Check if c is combination of a and b

$b = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$

$c = xa + yb \rightarrow c = -1(a) +$

$c = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$

$$\begin{cases} 2x + 3y = 1 & 2(-1) + 3y = 1, \; -2 + 3y = 1 \rightarrow y = 1 \\ -1x + 0y = 1 \rightarrow x = -1 \\ 2x + y = -1 \rightarrow 2(-1) + 1 = -1 \; \checkmark \text{ works} \end{cases}$$

$d = \begin{bmatrix} 5 \\ -1 \\ 3 \end{bmatrix}$

So $c = -a + b$, c is dependent on a, b

2. Check if d is a combination of a, b

$d = xa + yb$:

$$\begin{cases} 2x + 3y = 5 \Rightarrow 2(1) + 3y = 5 \Rightarrow 2 + 3y = 5, \; y = 1 \\ -x + 0y = -1 \Rightarrow x = 1 \\ 2x + y = 3 \Rightarrow 2(1) + 1 = 3 \; \checkmark \text{ works} \end{cases}$$

So $d = a + b$, d is dependent on a, b

3. Check if a, b are independent:

They're not multiples of each other (obvious by inspection; second components are -1 and 0, ratios differ in each coordinate), so they're linearly independent.

- All of c and d are in the span of $\{a, b\}$, so

$$T = \{a, b\} = \left\{ \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} \right\} \checkmark$$